

② Section A

- ① Machine
- ② Learning
- ③ Aspects
- ④ Finance
- ⑤ Criteria

Section B

- ① ⑥ It is raining and it is cold
- ② ⑥ Finds a substitution that makes two expressions identical
- ③ ⑥ Clause form
- ④ ⑥ First order logic
- ⑤ ⑥ An empty clause

Section C

- ① An Intelligent Agent is a system that perceives its environment through sensors and acts upon it using actuators to achieve specific goals.

Eg:-
A self-driving car is an intelligent agent
It uses sensors to understand traffic & hence make decision.

② Propositional Logic Predicate Logic

- | | |
|---|---|
| <p>① It deals with complete statement and does not consider internal structure
Eg. It is raining And it is cold</p> | <p>It deals with predicate, variables and objects and consider internal structure
Eg. Student (x) → Studies (x)</p> |
|---|---|

③ Training set -

In training set data is used to train the model and helps the model learn patterns

Test set - In test set data is used to evaluate the model and checks how well the model performs on new data

⊙ Avoid overfitting - (model memorizing instead of learning)

⊙ Test real ~~performance~~ Performance on unseen data

⊙ Ensure reliability of the model

⊙ Measure accuracy fairly

④ PCA (Principal Component Analysis) is a technique of feature (dimension) in a dataset while keeping most important information

It solves:

⊙ Reduce complexity of large dataset

⊙ Remove redundancy (duplicate or correlate data)

⊙ Speed up computation

⊙ Helps in visualization (eg. reducing to 2d or 3d)

⑤ Health care - AI help doctor diagnose disease and suggest treatment by analysing medical data

Finance - AI is used to detect fraud and manage investment by analysing transaction patterns

③ Robotics → AI enable robots to perform task automatically like manufacturing or delivery

Section 1

① Environment



Sensors



Performance Element



Actuators



Environment

Learning Element

(Improves performance)



Performance Element

②

Full Data set

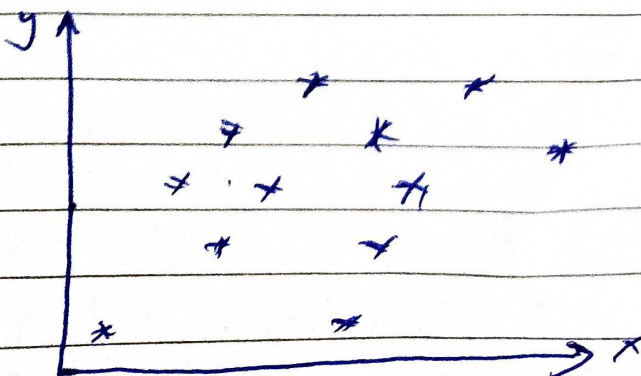


Training set

Validation set

Test set

③



PCI (new api)

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Bonus question

⇒ Advantage

AI can quickly analyse medical data & help doctors detect disease early

Limitation

AI may make error if data is incorrect or biased so it can't fully replace doctors